

# SQL SELECT Query

01 · Query Data — Complete Reference Guide

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## COMMENTS

→ *Single-line comment*

```
-- This is a single-line comment.
```

→ *Multi-line comment*

```
/* This  
  is  
  a multiple-line  
  comment  
*/
```

## 1. SELECT ALL COLUMNS

→ *Retrieve All Customer Data*

```
SELECT *  
FROM sales.customers
```

→ *Retrieve All Order Data*

```
SELECT *  
FROM sales.orders
```

## 2. SELECT SPECIFIC COLUMNS

→ *Retrieve each customer's name, country, and score*

```
SELECT  
    firstname,  
    country,  
    score  
FROM sales.customers
```

## 3. WHERE CLAUSE

→ *Retrieve customers with a score not equal to 0*

```
SELECT *  
FROM customers  
WHERE score != 0
```

→ *Retrieve customers from Germany*

```
SELECT *  
FROM customers  
WHERE country = 'Germany'
```

→ *Retrieve name and country of customers from Germany*

```
SELECT  
    first_name,  
    country  
FROM customers  
WHERE country = 'Germany'
```

## 4. ORDER BY

→ *Sort by highest score first (DESC)*

```
SELECT *  
FROM customers  
ORDER BY score DESC
```

→ *Sort by lowest score first (ASC)*

```
SELECT *  
FROM customers  
ORDER BY score ASC
```

→ *Sort by country*

```
SELECT *  
FROM customers  
ORDER BY country ASC
```

→ *Sort by country, then by highest score*

```
SELECT *  
FROM customers  
ORDER BY country ASC, score DESC
```

→ *Filter non-zero scores and sort by highest score*

```
SELECT  
    first_name,  
    country,  
    score  
FROM customers  
WHERE score != 0  
ORDER BY score DESC
```

## 5. GROUP BY

→ *Find the total score for each country*

```
SELECT  
    country,  
    SUM(score) AS total_score  
FROM customers  
GROUP BY country
```

■ Note: The query below will NOT work — 'first\_name' is neither in GROUP BY nor wrapped in an aggregate function.

```
SELECT  
    country,  
    first_name,          -- ERROR  
    SUM(score) AS total_score  
FROM customers  
GROUP BY country
```

→ *Find total score AND total number of customers per country*

```
SELECT  
    country,
```

```
SUM(score) AS total_score,  
COUNT(id) AS total_customers  
FROM customers  
GROUP BY country
```

## 6. HAVING

→ *Countries with average score greater than 430*

```
SELECT  
    country,  
    AVG(score) AS avg_score  
FROM customers  
GROUP BY country  
HAVING AVG(score) > 430
```

→ *Exclude zero scores, then filter countries with avg score > 430*

```
SELECT  
    country,  
    AVG(score) AS avg_score  
FROM customers  
WHERE score != 0  
GROUP BY country  
HAVING AVG(score) > 430
```

## 7. DISTINCT

→ *Return unique list of all countries*

```
SELECT DISTINCT country  
FROM customers
```

## 8. TOP

→ *Retrieve only 3 customers*

```
SELECT TOP 3 *  
FROM customers
```

→ *Top 3 customers with the highest scores*

```
SELECT TOP 3 *  
FROM customers  
ORDER BY score DESC
```

→ *Lowest 2 customers based on score*

```
SELECT TOP 2 *
```

```
FROM customers
ORDER BY score ASC
```

→ *Get the two most recent orders*

```
SELECT TOP 2 *
FROM orders
ORDER BY order_date DESC
```

## 9. Combining Queries — All Together

→ *Average score per country (non-zero), avg > 430, sorted highest first*

```
SELECT
    country,
    AVG(score) AS avg_score
FROM customers
WHERE score != 0
GROUP BY country
HAVING AVG(score) > 430
ORDER BY AVG(score) DESC
```

## 10. COOL STUFF — Additional SQL Features

→ *Execute multiple queries at once*

```
SELECT * FROM customers;
SELECT * FROM orders;
```

→ *Select static / constant values without a table*

```
SELECT 123 AS static_number;
SELECT 'Hello' AS static_string;
```

→ *Assign a constant value to a column in a query*

```
SELECT
    id,
    first_name,
    'New Customer' AS customer_type
FROM customers;
```